

EC614: Macroeconomic Forecasting - Final Exam

Multivariate Time Series Analysis

School of Economics, Quaid-i-Azam University

2026-01-12

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[EC614: Macroeconomic Forecasting
Final Examination]

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#text(size: 12pt)[Multivariate Time Series Analysis
School of Economics, Quaid-i-Azam University
January 12, 2026]]

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= Instructions

- *Time Allowed:* 3 hours
- *Total Marks:* 100
- *Instructor:* Prof. Dr. Zahid Asghar

== General Guidelines

- Answer ALL questions
- Show all working where applicable
- State any assumptions clearly
- Use economic intuition alongside technical analysis
- Diagrams and equations should be clearly labeled
- Write your answers in the space provided or on additional sheets if needed

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= Part A: Multiple Choice Questions (20 marks)

Each question carries 1 mark. Choose the BEST answer.

1. In a reduced-form VAR, the error terms across equations are:
 - (a) Always uncorrelated by construction
 - (b) May be correlated if variables are related
 - (c) Always normally distributed
 - (d) Independent of past values

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2. The primary advantage of ARDL over traditional cointegration methods is:

- (a) It requires all variables to be I(1)
- (b) It works with variables of mixed integration orders I(0) and I(1)
- (c) It is computationally simpler
- (d) It doesn't require lag selection

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3. In Granger causality testing, if X Granger-causes Y, it means:

- (a) X definitely causes Y in a causal sense
- (b) Past values of X help predict Y beyond Y's own past
- (c) Y cannot Granger-cause X simultaneously
- (d) X and Y must be cointegrated

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4. The Pesaran-Shin-Smith (2001) bounds test in ARDL tests the null hypothesis:

- (a) All variables are I(1)
- (b) Variables are cointegrated
- (c) No long-run relationship exists
- (d) Variables are stationary

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5. In a structural VAR (SVAR), identification restrictions are imposed on:

- (a) The lag length only

- (b) The contemporaneous relationships among variables
- (c) The forecast horizon
- (d) The variance of residuals

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6. Which statement about impulse response functions (IRFs) is FALSE?

- (a) They trace dynamic effects of shocks over time
- (b) They require the VAR to have uncorrelated errors
- (c) Confidence bands indicate statistical uncertainty
- (d) They are always symmetric between variables

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7. In a recursive VAR, the ordering of variables matters because:

- (a) It determines computational efficiency
- (b) It imposes assumptions about contemporaneous effects
- (c) It affects lag length selection
- (d) It changes the sample size

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8. The forecast error variance decomposition (FEVD) answers:

- (a) How accurate the forecast is
- (b) What fraction of forecast error variance is due to each shock
- (c) Whether the model is correctly specified
- (d) If variables are cointegrated

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9. If two I(1) variables are cointegrated, their linear combination is:

- (a) I(1)
- (b) I(2)
- (c) I(0)
- (d) Non-stationary with no defined order

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10. The error correction term in a VECM represents:

- (a) The forecasting error
- (b) Deviation from long-run equilibrium
- (c) Short-run dynamics
- (d) Measurement error

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11. In the Johansen cointegration test, the trace statistic tests:

- (a) Whether $\text{rank}(\Pi) = 0$
- (b) Whether $\text{rank}(\Pi) \leq r$ against $\text{rank}(\Pi) > r$
- (c) Stationarity of individual series
- (d) Granger causality

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12. Weak exogeneity in a VECM context means:

- (a) The variable doesn't respond to disequilibrium
- (b) The variable is determined outside the model
- (c) The variable is stationary
- (d) The variable has weak predictive power

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13. The primary problem with estimating a VAR with non-stationary I(1) variables that are NOT cointegrated is:

- (a) Computational complexity
- (b) Spurious regression
- (c) Loss of degrees of freedom
- (d) Heteroskedasticity

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14. In ARDL modeling, if the F-statistic falls between the lower and upper bounds:

- (a) We reject cointegration
- (b) We accept cointegration
- (c) The result is inconclusive
- (d) We need to increase lag length

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15. The Cholesky decomposition in recursive VAR estimation:

- (a) Tests for cointegration
- (b) Orthogonalizes the VAR residuals
- (c) Selects optimal lag length
- (d) Estimates structural parameters

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16. Compared to univariate ARIMA models, VARs:

- (a) Are always superior for forecasting
- (b) Capture cross-variable dynamics
- (c) Require less data
- (d) Have clearer structural interpretation

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17. In panel data with fixed effects, the “within” transformation:

- (a) Adds time-specific effects
- (b) Removes individual-specific means
- (c) Differences all variables
- (d) Standardizes variables

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18. The Hausman test in panel data compares:

- (a) Fixed effects vs pooled OLS
- (b) Random effects vs pooled OLS
- (c) Fixed effects vs random effects
- (d) Static vs dynamic models

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19. In SVAR identification using the Blanchard-Perotti approach:

- (a) Only short-run restrictions are used
- (b) Institutional knowledge provides parameter values
- (c) All variables must be ordered recursively
- (d) Long-run restrictions are necessary

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20. When VAR residuals show autocorrelation, the most likely issue is:

- (a) Too many lags included
- (b) Insufficient lag length
- (c) Non-stationarity
- (d) Heteroskedasticity

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= Part B: Short Questions (30 marks)

Answer ALL questions. Each question carries 10 marks.

== Question 1 (10 marks)

Explain the difference between reduced-form VAR, recursive VAR, and structural VAR. Under what circumstances would you prefer using each type? Provide a concrete example from macroeconomic policy analysis.

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== Question 2 (10 marks)

You estimate an ARDL(2,4,0,0,0,4,4,1) model for a relationship between multiple variables.

(a) What do the numbers in parentheses represent? (3 marks)

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(b) The bounds test F-statistic = 6.45 (critical values: $I(0) = 2.45$, $I(1) = 3.61$ at 5% level). What do you conclude about cointegration? (3 marks)

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(c) If cointegration exists, what is your next step in the analysis? (4 marks)

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== Question 3 (10 marks)

Consider a three-variable VECM for money (M), prices (P), and GDP (Y) for Pakistan.

The Johansen test indicates $r = 1$ cointegrating relationship. The estimated cointegrating vector is:

$\beta' = [1, -0.85, -0.35]$ for $[M, P, Y]$

(a) Interpret this long-run relationship economically.
(5 marks)

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(b) What does $r = 1$ tell you about the system? (3 marks)

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(c) How would you test if money (M) is weakly exogenous? (2 marks)

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= Part C: Long Questions (50 marks)

Answer ALL questions.

== Question 4 (25 marks)

You are analyzing monetary policy transmission in Pakistan using quarterly data from 2000Q1 to 2023Q4 for the following variables:

- GDP growth rate (Y)
- CPI inflation (π)
- Policy interest rate (R)
- Real effective exchange rate (REER)

=== Part (a): Preliminary Analysis (8 marks)

Answer the following:

(i) Explain the steps you would take to prepare the data before estimating a VAR model. (3 marks)

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(ii) What unit root tests would you apply and why? (3 marks)

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(iii) How would you handle the different orders of integration if some variables are $I(0)$ and others are $I(1)$? (2 marks)

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=== Part (b): VAR Specification (9 marks)

Answer the following:

(i) Explain how you would select the optimal lag length for the VAR. (3 marks)

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(ii) Would you include deterministic terms (constant, trend)? Justify your answer. (3 marks)

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(iii) Write out the VAR equations in matrix form for lag order $p=2$. (3 marks)

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=== Part (c): Structural Identification (8 marks)

Answer the following:

(i) Propose a recursive ordering for these variables based on institutional knowledge of Pakistan's economy. Justify your ordering. (3 marks)

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(ii) Alternatively, propose one short-run restriction and one long-run restriction for SVAR identification. (3 marks)

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(iii) What economic shock are you trying to identify, and why does it matter for policy? (2 marks)

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== Question 5 (25 marks)

A researcher estimates the following VECM for consumption (C), income (Y), and wealth (W) using Pakistani data:

Long-run relationship (cointegrating equation):

$$C_t = 0.75 Y_t + 0.15 W_t$$

Short-run dynamics:

$$\Delta C_t = -0.25(C_{t-1} - 0.75 Y_{t-1} - 0.15 W_{t-1}) + 0.40 \Delta Y_t + 0.10 \Delta W_t$$

$$\Delta Y_t = 0.05(C_{t-1} - 0.75 Y_{t-1} - 0.15 W_{t-1}) + 0.65 \Delta Y_{t-1}$$

$$\Delta W_t = -0.02(C_{t-1} - 0.75 Y_{t-1} - 0.15 W_{t-1}) + 0.25 \Delta Y_t + 0.50 \Delta W_{t-1}$$

=== Part (a): Long-run Interpretation (10 marks)

(i) What is the long-run marginal propensity to consume out of income and wealth? (3 marks)

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(ii) Is this relationship consistent with economic theory? Explain. (4 marks)

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(iii) What does the cointegrating relationship tell you about the permanent income hypothesis? (3 marks)

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=== Part (b): Short-run Dynamics (10 marks)

(i) Interpret the error correction coefficients (-0.25, 0.05, -0.02). Which variable(s) adjust to disequilibrium? (4 marks)

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(ii) Is wealth (W) weakly exogenous? Explain the implications. (3 marks)

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(iii) What is the immediate (impact) effect of a 1% increase in income on consumption? (3 marks)

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=== Part (c): Policy Implications (5 marks)

(i) If the government wants to stimulate consumption through fiscal policy, should it focus on increasing current income or wealth? Use the estimates to support your answer. (3 marks)

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(ii) What are the limitations of this VECM for policy analysis in Pakistan's context? (2 marks)

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= Marking Scheme Summary

#table(columns: 4, align: (left, left, center, center),
[Section], [Questions], [Marks per Question], [Total
Marks], [Part A], [20 MCQs], [1], [20], [Part B], [3

Short Questions], [10], [30], [Part C], [2 Long Questions], [25], [50], [TOTAL], [], [], [100])

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Examination]

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#text(size: 12pt)[Good Luck!]]